Viscon Dua Programming and Diagnostic Utility (PDU)

Configuration Manual for FTU

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REVISION CHANGE HISTORY

Revision	Date	Prepared by	Pages Affected	Comment

LIST OF APPENDICES

Appendix	Title

GLOSSARY

Abbreviation/Term	Description
CAASDU	Common Address ASDU
CES	Control Enable Switch
COT	Cause Of Transmission
CPU	Central Processing Unit
FTU	Field Terminal Unit
IEC	International Electrotechnical Commission
IOA	Information Object Address
IP	Internet Protocol
LA	Link Address
OA	Originator Address
PDU	Programming and Diagnostic Utility
RTU	Remote Terminal Unit
SCADA	Supervisory Control and Data Acquisitions
TCP	Transmission Control Protocol

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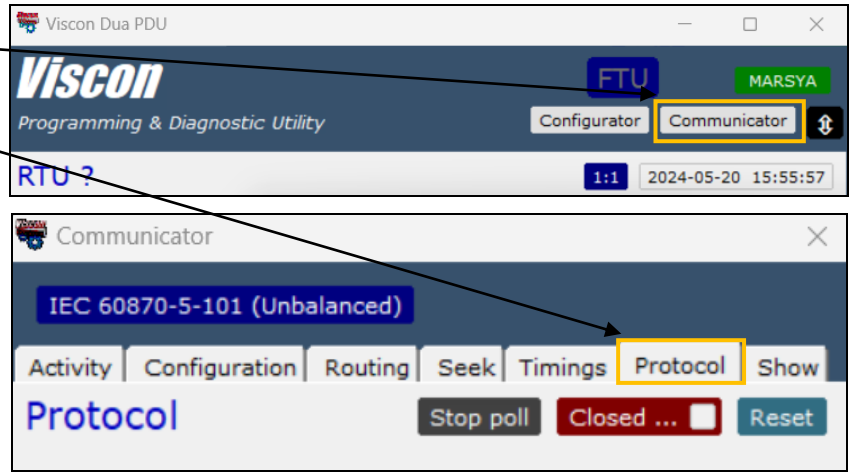
1. COMMUNICATION TEST

1.1 Communication Setup and Polling

1.1.1 Objectives

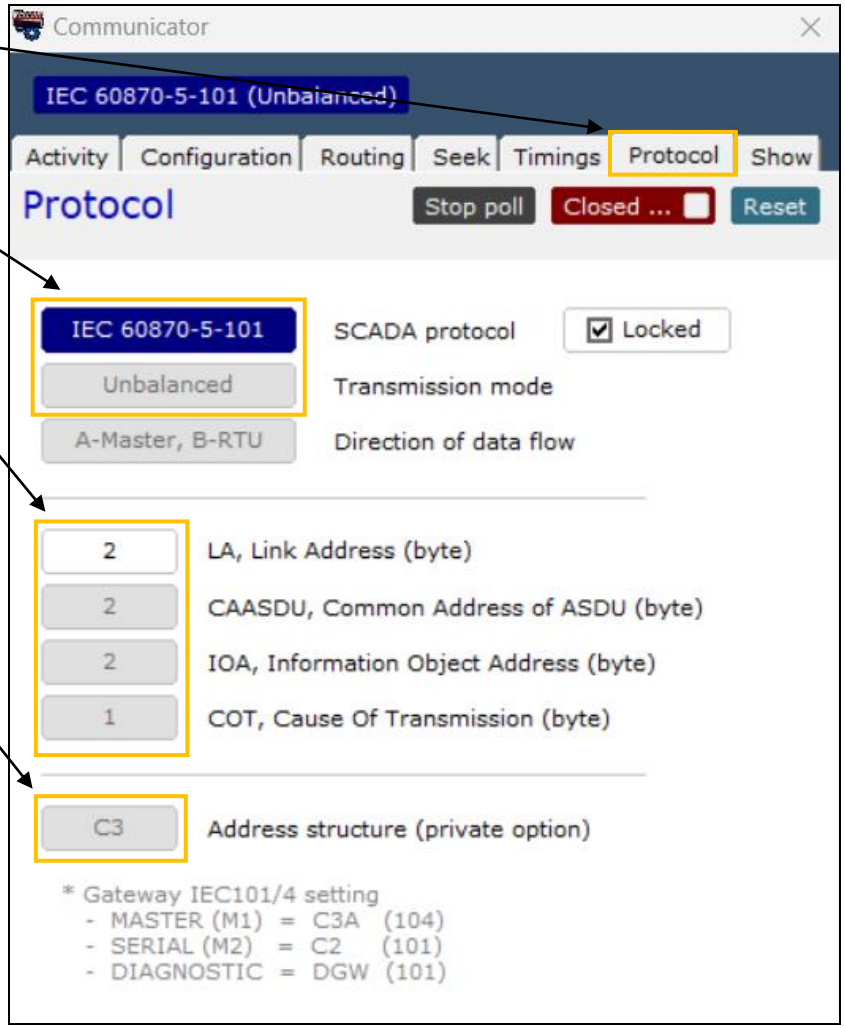
To demonstrate the ability of the PDU to communicate with the set-up device.

1.1.2 Procedure

1. Select Communicator > Protocol.	

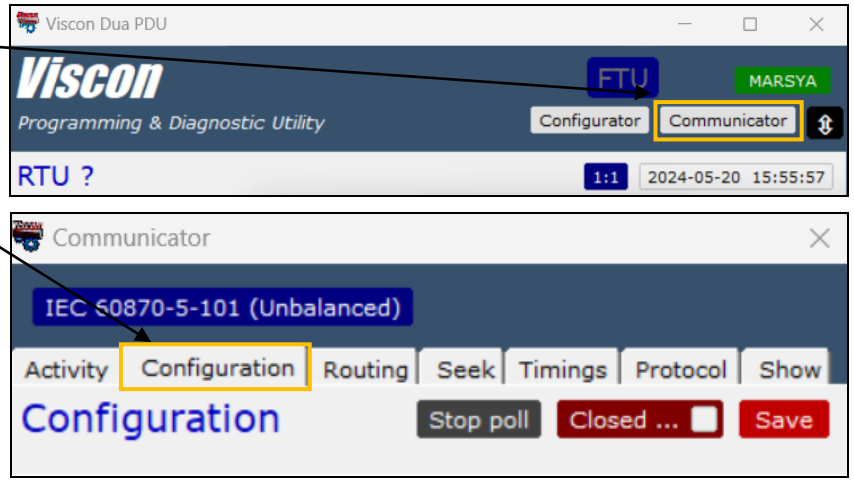
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2. In the Protocol tab, set each parameter as below:
   a. Protocol: **IEC 60870-5-101**
   b. Transmission mode: **Unbalanced**
   c. Link Address (LA) byte size: **2**
   d. Common Address ASDU (CAASDU) byte size: **2**
   e. Information Object Address (IOA) byte size: **2**
   f. Cause Of Transmission (COT) byte size: **1**
   g. Address structure (Private option): **C3**



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3. Select Communicator > Configuration.



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4. In the Configuration tab, set each parameter as below:

   Using serial:

   a. Serial 1 (RS-232): **COM3** (Refer to Device Manager > Port (COM & LPT))

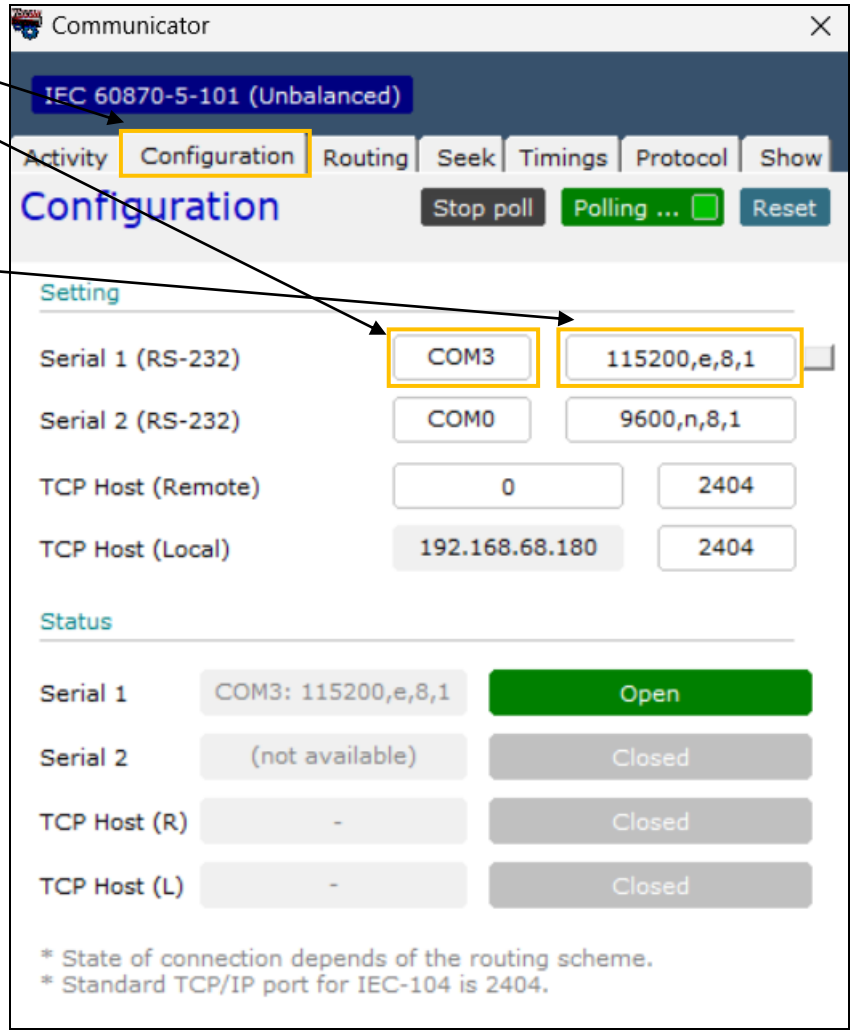
   b. Speed: **115200**

   c. Parity: **e**

   d. Size: **8**

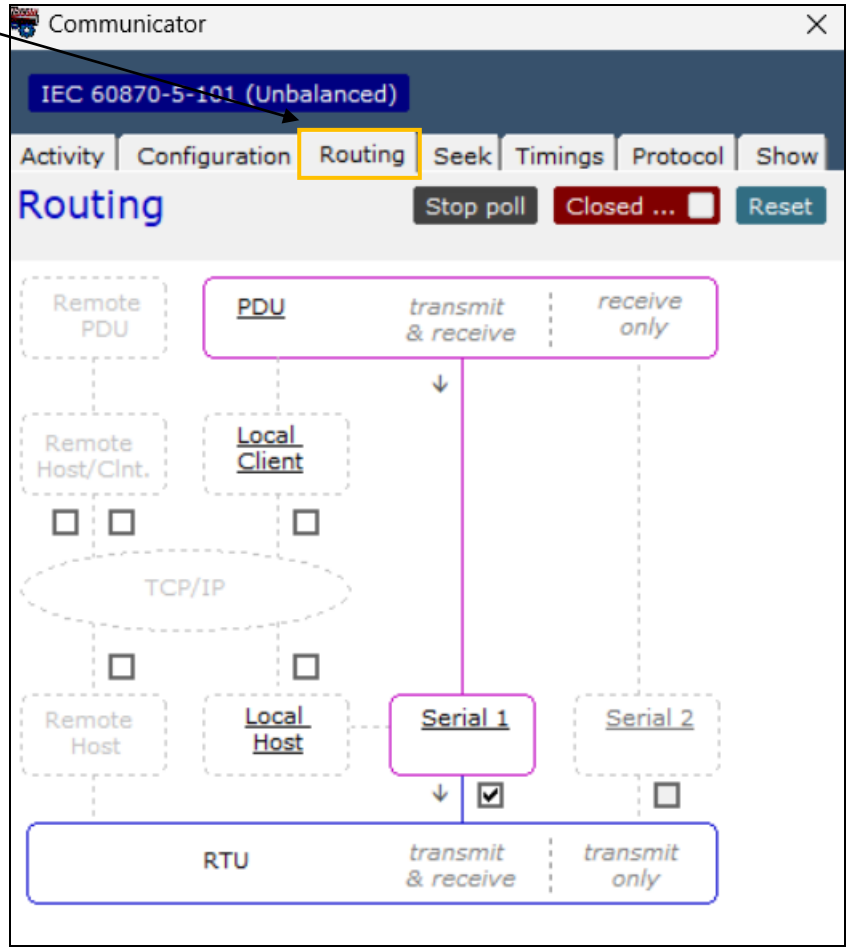
   e. Stopbit: **1**

   Refer to RTU module settings



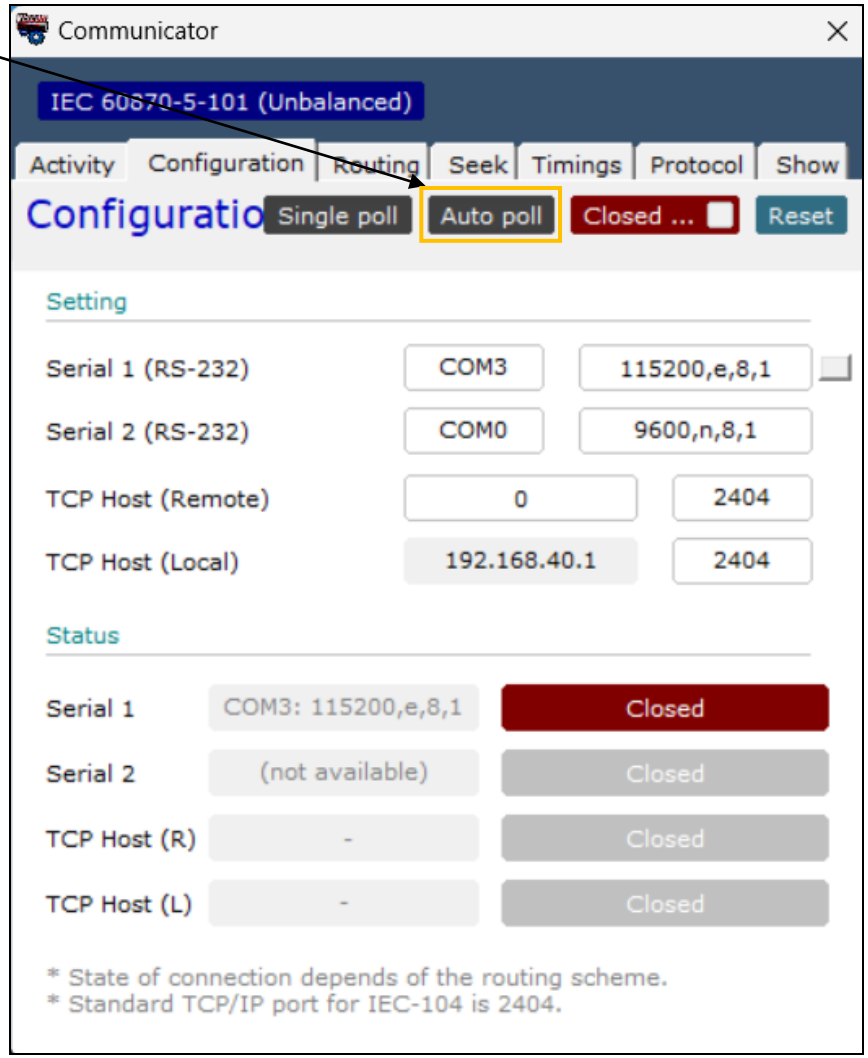
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5. In the Routing tab, select the route PDU through Serial 1 to RTU.



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6. Then, select auto poll to initiate polling with set up device.



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2. CONFIGURE PARAMETERS TEST

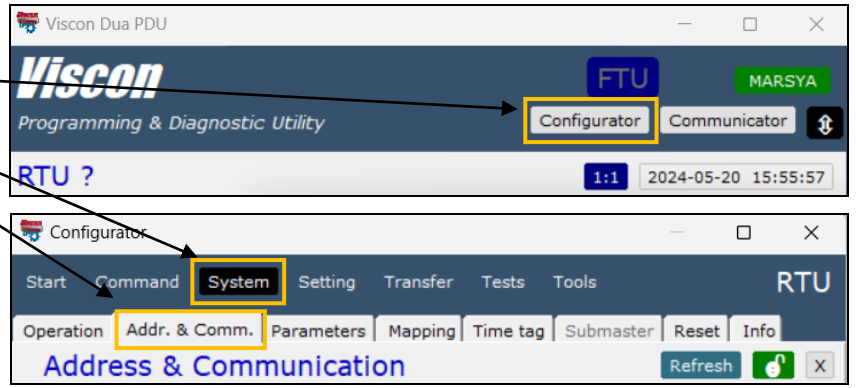
2.1 Set Device's Link Address (LA) and Common ASDU (CAASDU)

2.1.1 Objectives

To demonstrate the ability of the PDU to set device's link address and common ASDU.

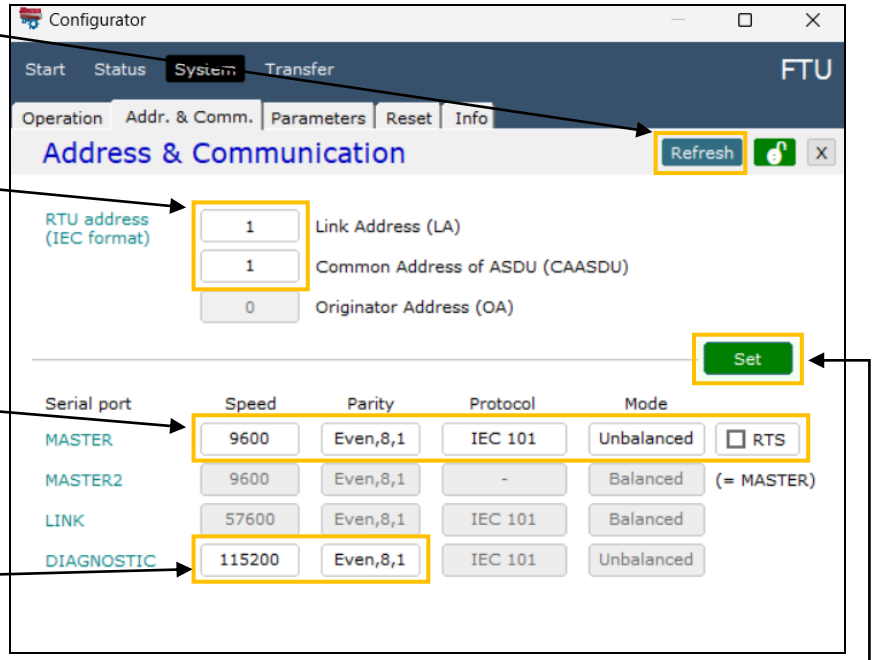
2.1.2 Procedure

1. Set PDU to communicate with the device.
2. Select Configurator > System > Addr. & Comm.



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3. Click on Refresh button to retrieve information on the device.
4. On RTU address (IEC format), set each parameters according to designated product. Example as below:
   a. Link Address (LA) = **1**
   b. Common Address of ASDU (CAASDU) = **1**
5. On Serial port: MASTER set:
   a. Speed = **9600**
   b. Parity = **Even,8,1**
   c. Protocol = **IEC 101**
   d. Mode = **Unbalanced**
   e. RTS = **Unchecked**
6. On Serial port: DIAGNOSTIC set:
   a. Speed = **115200**
   b. Parity = **Even,8,1**
7. Other parameters if not listed above shall remain as is or default value.
8. Click Set to save the configuration into the device.
9. Click Refresh to verify the settings has been set on the device. The refresh is used to extract data from the designated device/RTU.



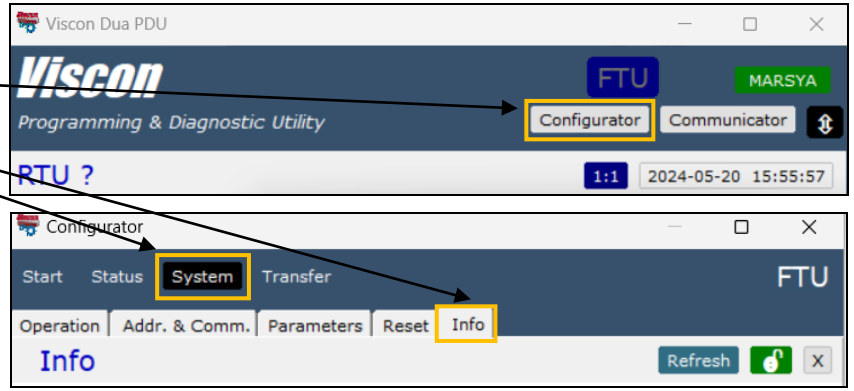
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2.2 Check Device's Info

2.2.1 Objectives

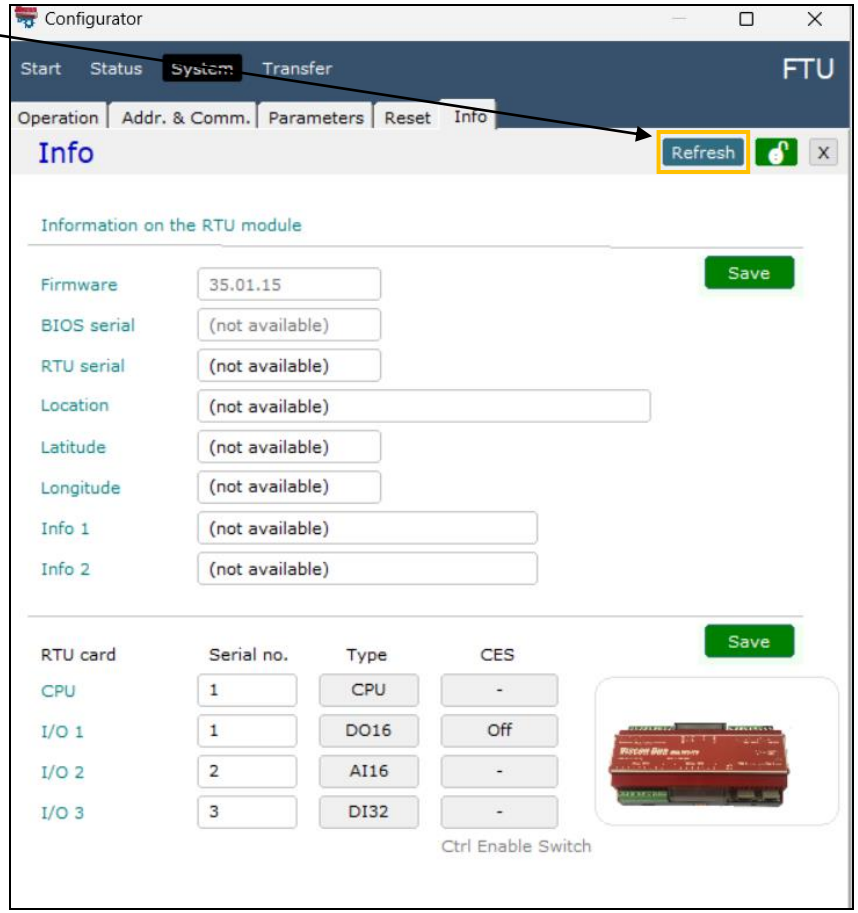
To demonstrate the ability of the PDU to retrieve device's info.

2.2.2 Procedure

1. Set PDU to communicate with the device. 2. Select Configurator > System > Info.	

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3. Click on Refresh button to retrieve information on the device.
4. Info provided by the device will display on the respective page



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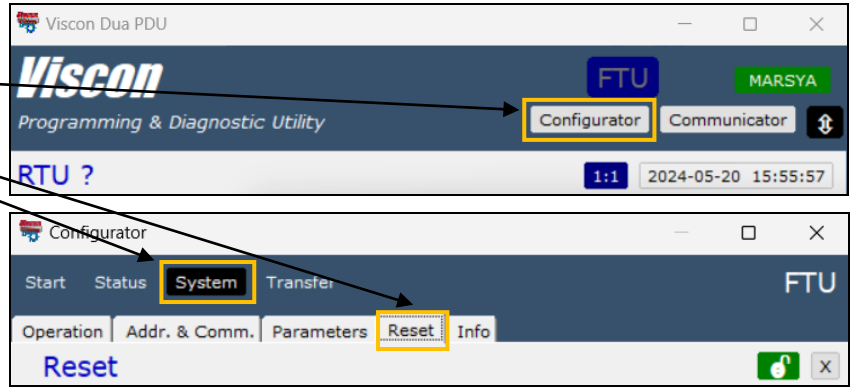
2.3 Reset Device's Configuration

2.3.1 Objectives

To demonstrate the ability of the PDU to reset device's configuration.

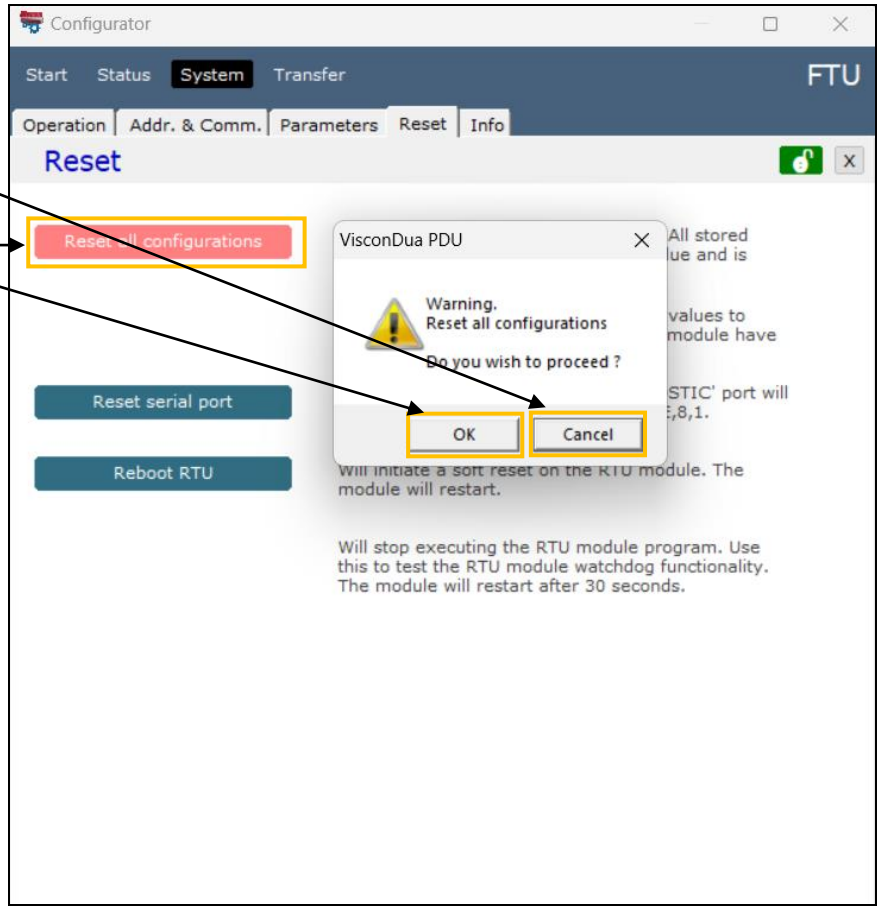
2.3.2 Procedure

1. Set PDU to communicate with the device.
2. Select Configurator > System > Reset.



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3. Click on Reset all configurations.
4. Click OK to reset all configurations or Cancel to cancel/terminate the action.
5. This action will re-initialise the configuration files. All stored parameters will be set to its default value and is irreversible.



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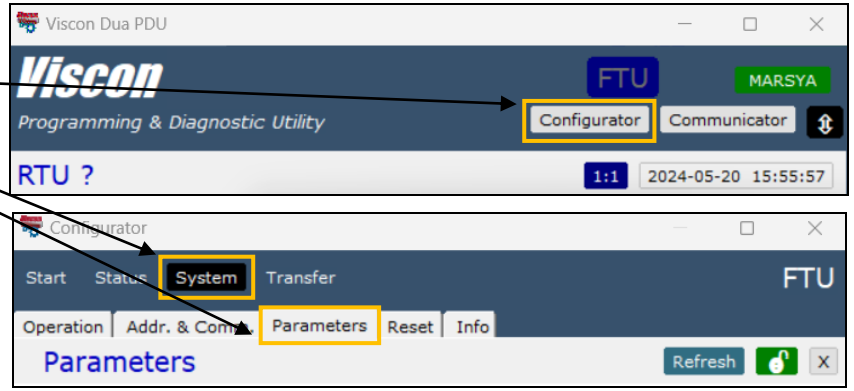
2.4 Set Device's Parameters

2.4.1 Objectives

To demonstrate the ability of the PDU to set device's parameters.

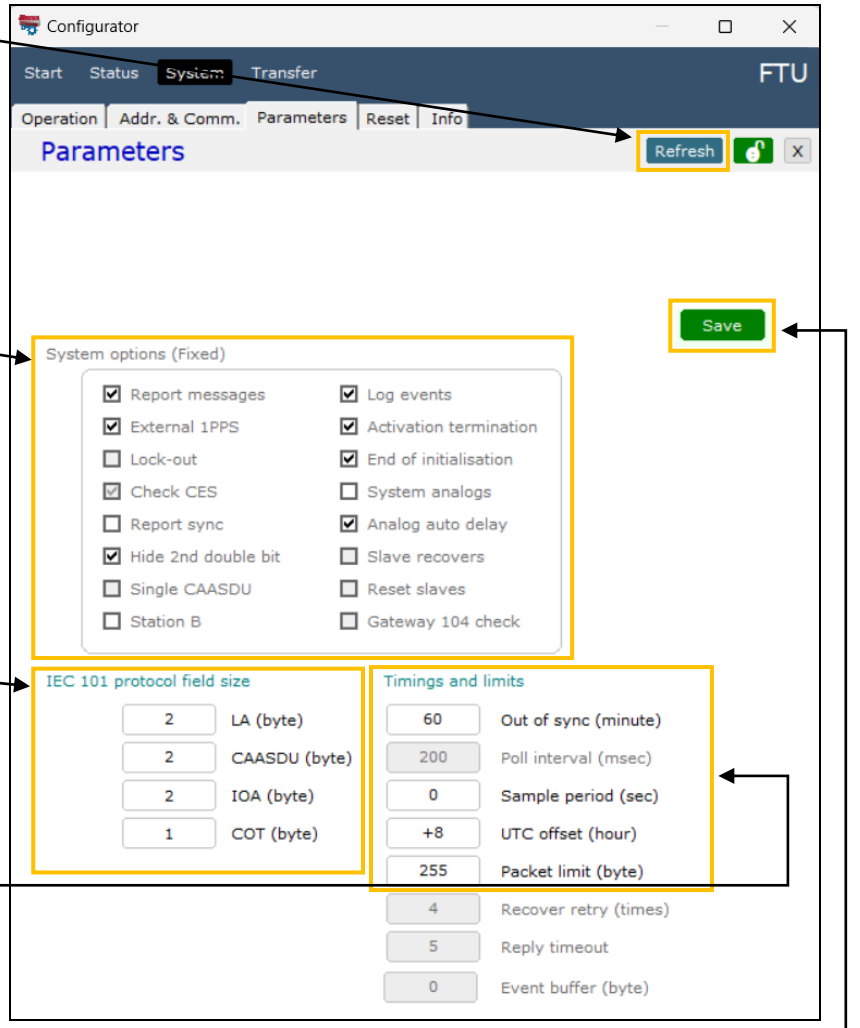
2.4.2 Procedure

1. Set PDU to communicate with the device.
2. Select Configurator > System > Parameters.



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3. Click on Refresh button to retrieve information on the device.
4. On System options, set the system's setting:
   a. Report messages
   b. External 1PPS
   c. Check CES
   d. Hide 2nd double bit
   e. Log events
   f. Activation termination
   g. End of initialisation
   h. Analog auto delay
5. Click and insert value for designated/targeted IEC 101 protocol field size. Example as below:
   a. LA (byte) = **2**
   b. CAASDU (byte) = **2**
   c. IOA (byte) = **2**
   d. COT (byte) = **1**
6. Click and insert value for Timings and limits:
   a. Out of sync (minute) = **60**
   b. Sample period (sec) = **0**
   c. UTC offset (hour) = **+8**
   d. Packet limit (byte) = **255**
7. Other parameters if not listed above shall remain as is or default value.
8. Click Save to save the configuration into the device.



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9. Click Refresh to verify the settings has been set at the device.	

2.5 Set Device's Time Tag

The "Time Tag" option is not visible in the settings menu because the settings has already been fixed.

2.6 Set Device's Digital Input

The "Digital Input" option is not visible in the settings menu because the settings has already been fixed.

2.7 Set Device's Digital Output

The "Digital Output" option is not visible in the settings menu because the settings has already been fixed.

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3. FILE TRANSFER TEST

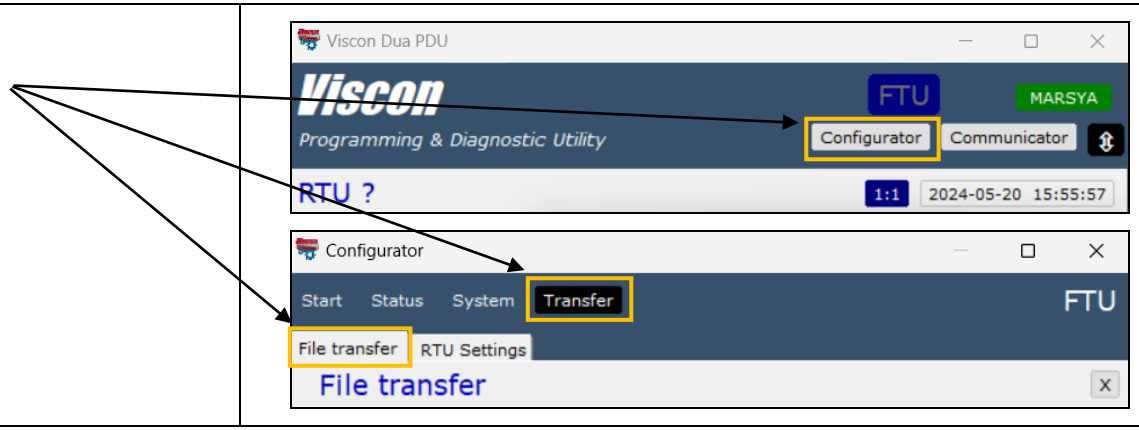
3.1 Download Device's Log Files

3.1.1 Objectives

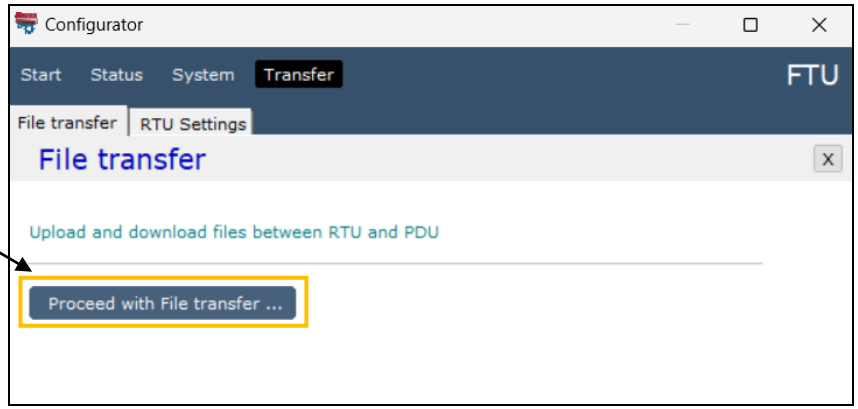
To demonstrate the ability of the PDU to transfer file from the device to the PDU local machine. (Download SOE)

3.1.2 Procedure

1. Set PDU to communicate with the device.
2. Select Configurator > Transfer > File Transfer.

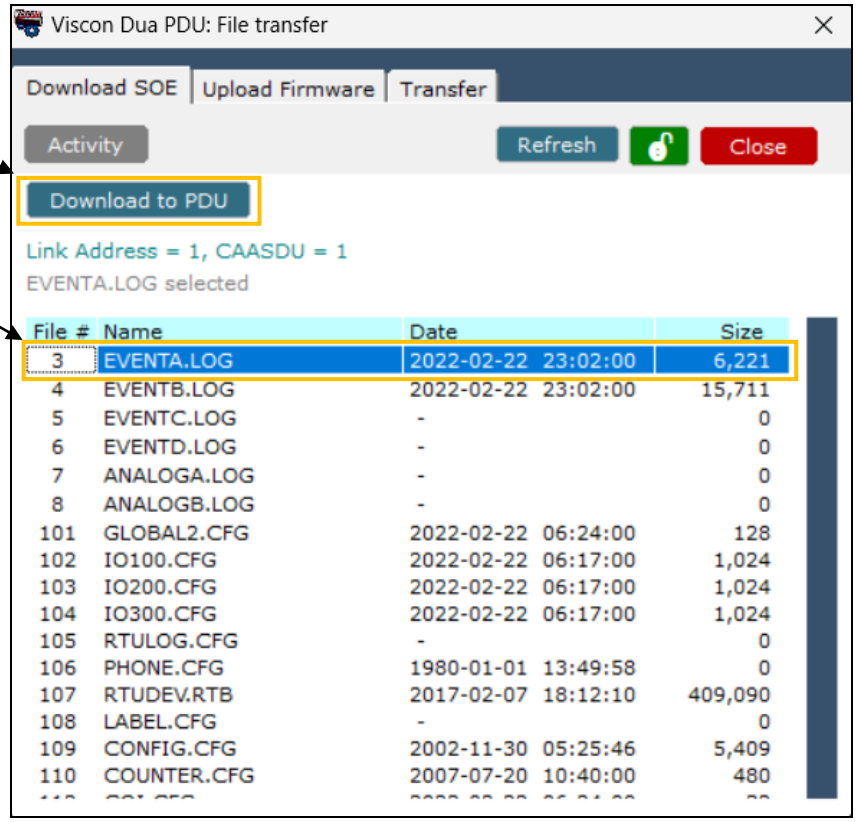


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3. Click on Proceed with File Transfer.	

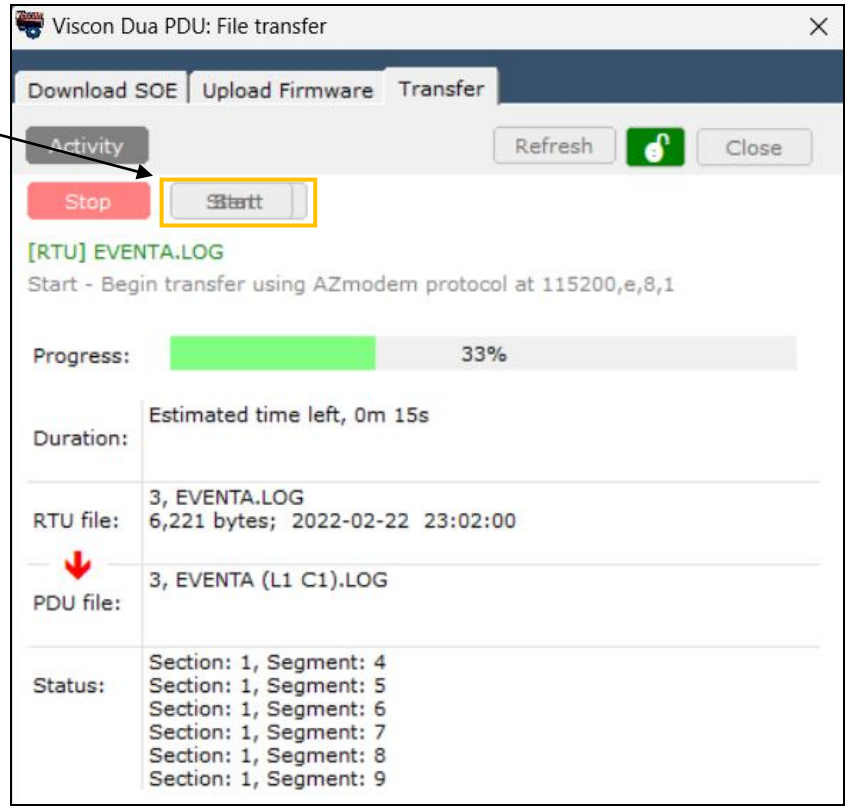
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4. In RTU files tab, select EVENTA.LOG, observe the file date to be compare after to download and click Download to PDU.

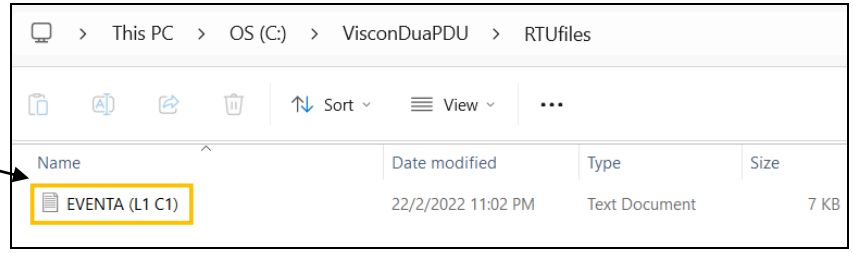


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5. In the Transfer tab, click Start to begin downloading EVENTA.LOG.

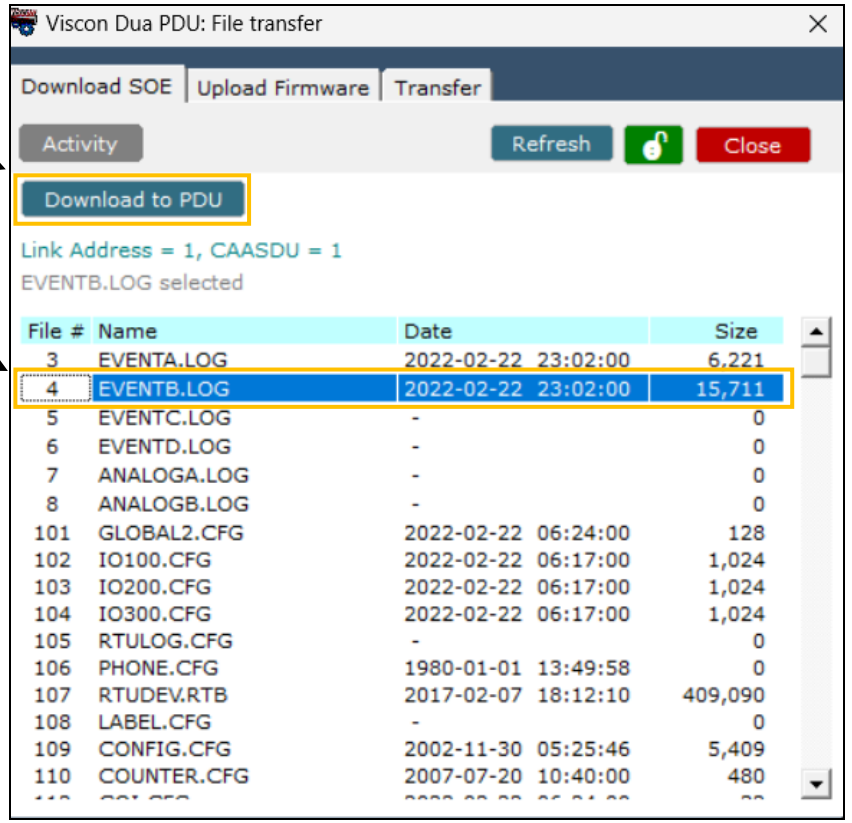


6. The EVENTA.LOG file is downloaded and located at C:\ViscconDuaPDU\RTUfiles.



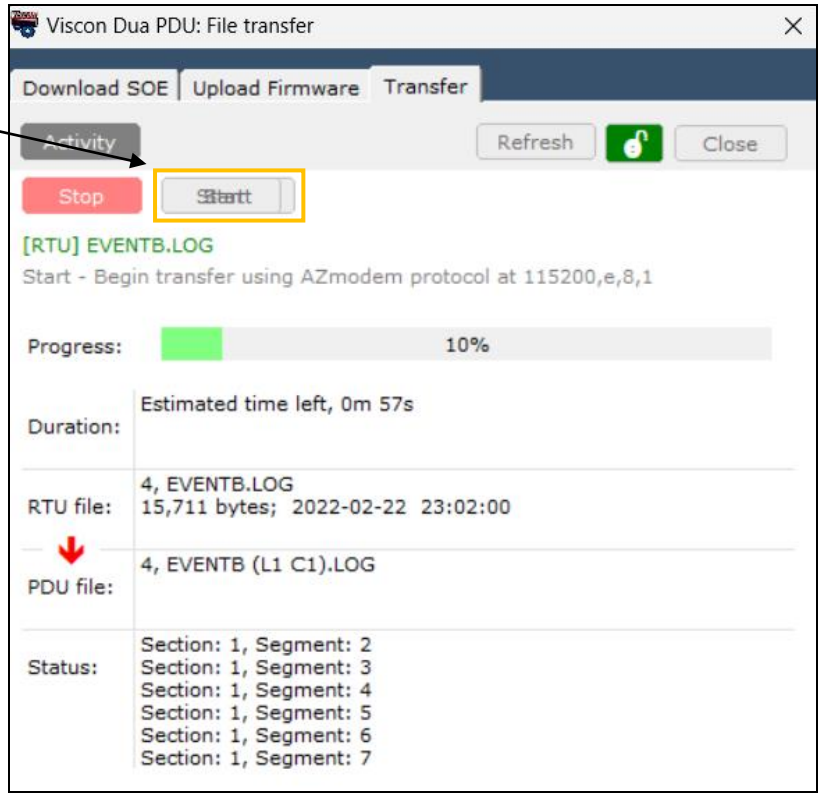
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7. Then select EVENTB.LOG, observe the file date to be compare after to download and click Download to PDU.

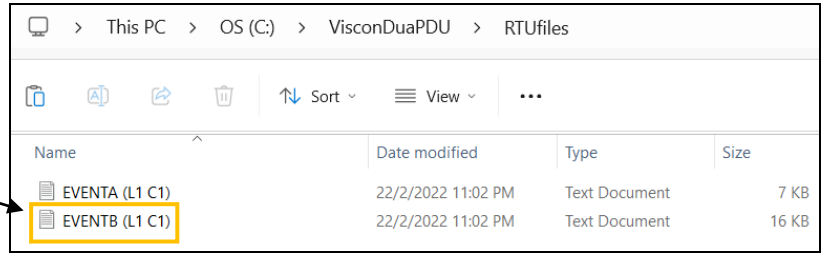


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8. In the Transfer tab, click Start to begin downloading EVENTB.LOG.

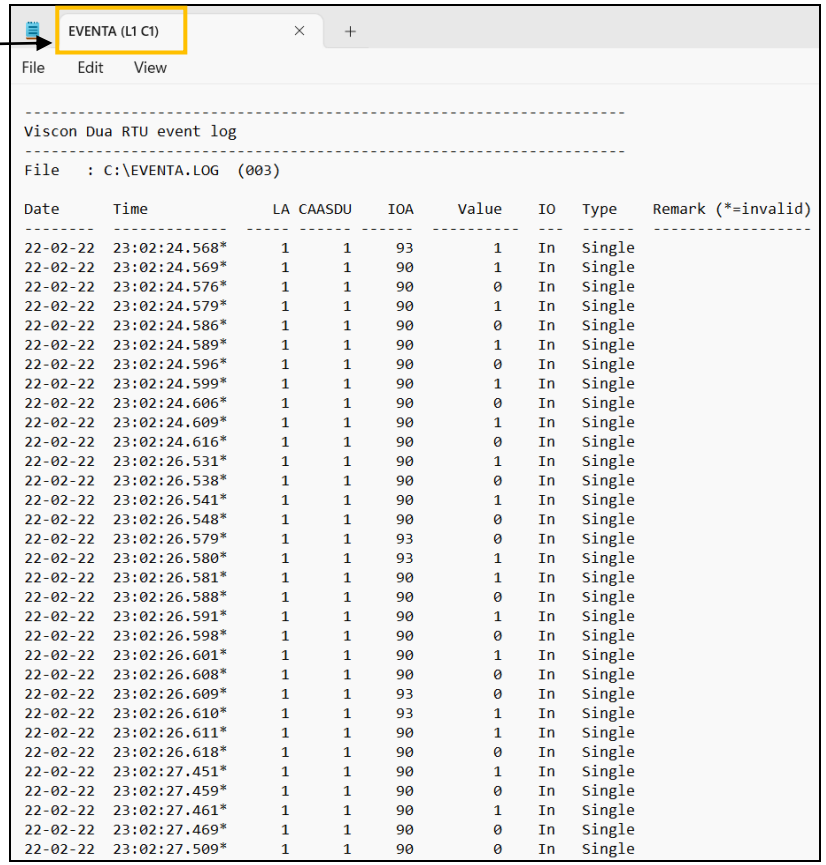


9. The EVENTB.LOG file is downloaded and located at C:\VisconDuaPDU\RTUfiles.



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10. Open the EVENTA.LOG files using Notepad application to verify the content exists.

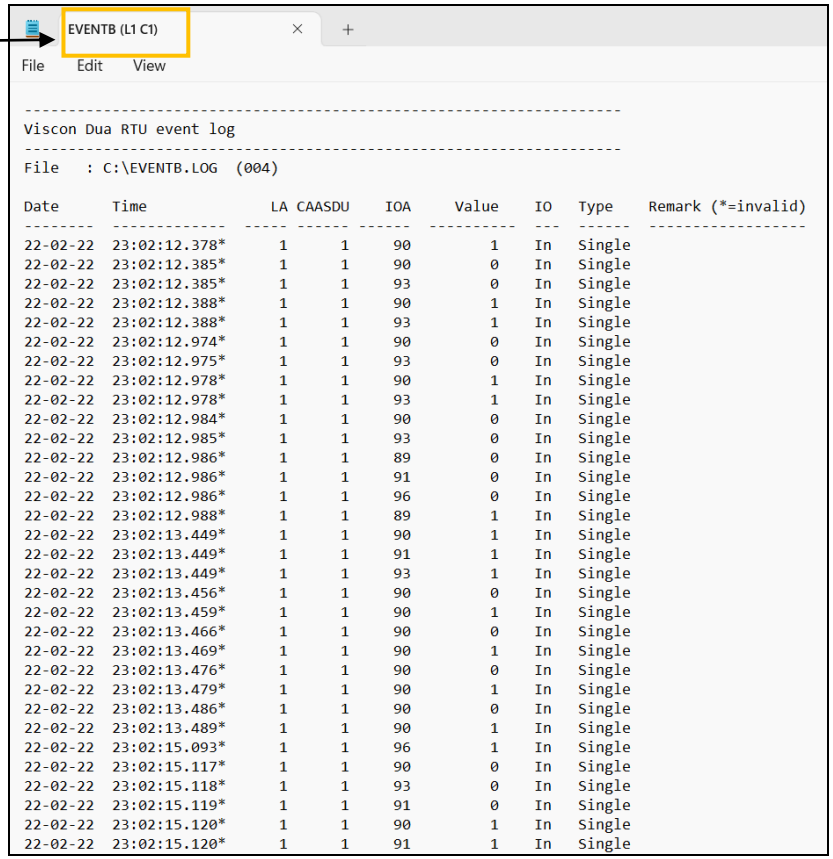



```
------------------------------------------------------------------------
Viscon Dua RTU event log
------------------------------------------------------------------------
File   : C:\EVENTA.LOG  (003)

Date      Time             LA CAASDU    IOA     Value    IO   Type    Remark (*=invalid)
--------  -------------  ----- ------ ------  ----------  ---  ------  ------------------
22-02-22  23:02:24.568*      1      1     93           1  In   Single
22-02-22  23:02:24.569*      1      1     90           1  In   Single
22-02-22  23:02:24.576*      1      1     90           0  In   Single
22-02-22  23:02:24.579*      1      1     90           1  In   Single
22-02-22  23:02:24.586*      1      1     90           0  In   Single
22-02-22  23:02:24.589*      1      1     90           1  In   Single
22-02-22  23:02:24.596*      1      1     90           0  In   Single
22-02-22  23:02:24.599*      1      1     90           1  In   Single
22-02-22  23:02:24.606*      1      1     90           0  In   Single
22-02-22  23:02:24.609*      1      1     90           1  In   Single
22-02-22  23:02:24.616*      1      1     90           0  In   Single
22-02-22  23:02:26.531*      1      1     90           1  In   Single
22-02-22  23:02:26.538*      1      1     90           0  In   Single
22-02-22  23:02:26.541*      1      1     90           1  In   Single
22-02-22  23:02:26.548*      1      1     90           0  In   Single
22-02-22  23:02:26.579*      1      1     93           0  In   Single
22-02-22  23:02:26.580*      1      1     93           1  In   Single
22-02-22  23:02:26.581*      1      1     90           1  In   Single
22-02-22  23:02:26.588*      1      1     90           0  In   Single
22-02-22  23:02:26.591*      1      1     90           1  In   Single
22-02-22  23:02:26.598*      1      1     90           0  In   Single
22-02-22  23:02:26.601*      1      1     90           1  In   Single
22-02-22  23:02:26.608*      1      1     90           0  In   Single
22-02-22  23:02:26.609*      1      1     93           0  In   Single
22-02-22  23:02:26.610*      1      1     93           1  In   Single
22-02-22  23:02:26.611*      1      1     90           1  In   Single
22-02-22  23:02:26.618*      1      1     90           0  In   Single
22-02-22  23:02:27.451*      1      1     90           1  In   Single
22-02-22  23:02:27.459*      1      1     90           0  In   Single
22-02-22  23:02:27.461*      1      1     90           1  In   Single
22-02-22  23:02:27.469*      1      1     90           0  In   Single
22-02-22  23:02:27.509*      1      1     90           0  In   Single
```


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11. Open the EVENTB.LOG files using Notepad application to verify the content exists.




```
------------------------------------------------------------------------
Viscon Dua RTU event log
------------------------------------------------------------------------
File   : C:\EVENTB.LOG  (004)

Date      Time              LA CAASDU    IOA     Value    IO   Type    Remark (*=invalid)
--------  ------------  ----- ------ ------  ----------  ---  ------  ------------------
22-02-22  23:02:12.378*      1      1     90           1   In   Single
22-02-22  23:02:12.385*      1      1     90           0   In   Single
22-02-22  23:02:12.385*      1      1     93           0   In   Single
22-02-22  23:02:12.388*      1      1     90           1   In   Single
22-02-22  23:02:12.388*      1      1     93           1   In   Single
22-02-22  23:02:12.974*      1      1     90           0   In   Single
22-02-22  23:02:12.975*      1      1     93           0   In   Single
22-02-22  23:02:12.978*      1      1     90           1   In   Single
22-02-22  23:02:12.978*      1      1     93           1   In   Single
22-02-22  23:02:12.984*      1      1     90           0   In   Single
22-02-22  23:02:12.985*      1      1     93           0   In   Single
22-02-22  23:02:12.986*      1      1     89           0   In   Single
22-02-22  23:02:12.986*      1      1     91           0   In   Single
22-02-22  23:02:12.986*      1      1     96           0   In   Single
22-02-22  23:02:12.988*      1      1     89           1   In   Single
22-02-22  23:02:13.449*      1      1     90           1   In   Single
22-02-22  23:02:13.449*      1      1     91           1   In   Single
22-02-22  23:02:13.449*      1      1     93           1   In   Single
22-02-22  23:02:13.456*      1      1     90           0   In   Single
22-02-22  23:02:13.459*      1      1     90           1   In   Single
22-02-22  23:02:13.466*      1      1     90           0   In   Single
22-02-22  23:02:13.469*      1      1     90           1   In   Single
22-02-22  23:02:13.476*      1      1     90           0   In   Single
22-02-22  23:02:13.479*      1      1     90           1   In   Single
22-02-22  23:02:13.486*      1      1     90           0   In   Single
22-02-22  23:02:13.489*      1      1     90           1   In   Single
22-02-22  23:02:15.093*      1      1     96           1   In   Single
22-02-22  23:02:15.117*      1      1     90           0   In   Single
22-02-22  23:02:15.118*      1      1     93           0   In   Single
22-02-22  23:02:15.119*      1      1     91           0   In   Single
22-02-22  23:02:15.120*      1      1     90           1   In   Single
22-02-22  23:02:15.120*      1      1     91           1   In   Single
```


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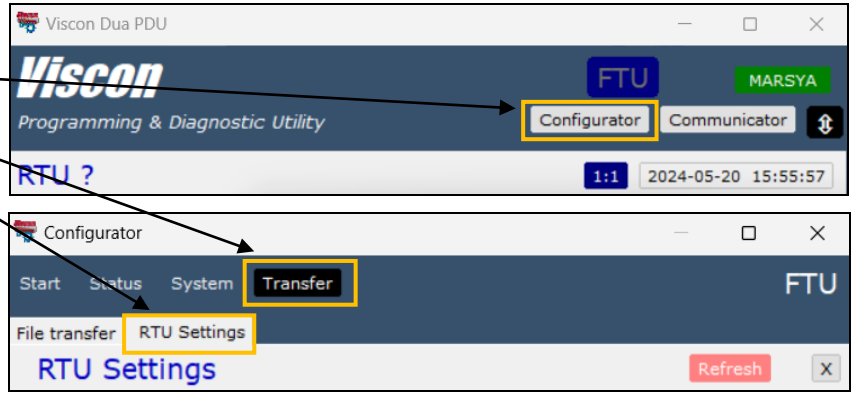
3.2 Download Device's Settings

3.2.1 Objectives

To demonstrate the ability of the PDU to transfer settings from the device to the PDU local machine.

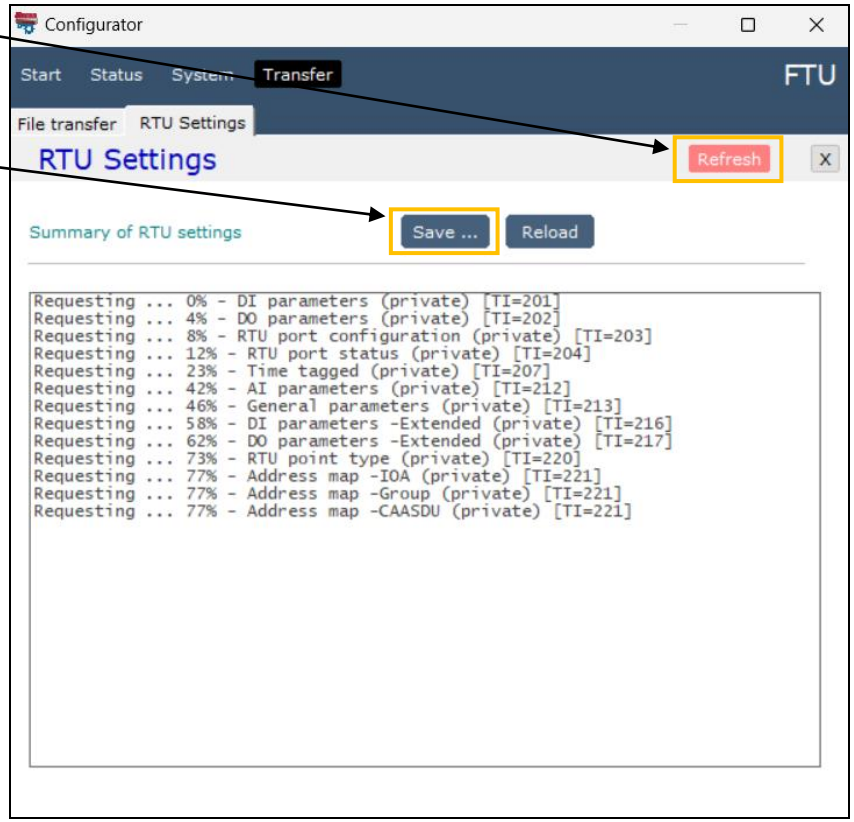
3.2.2 Procedure

1. Set PDU to communicate with the device.
2. Select Configurator > Transfer > RTU Settings.



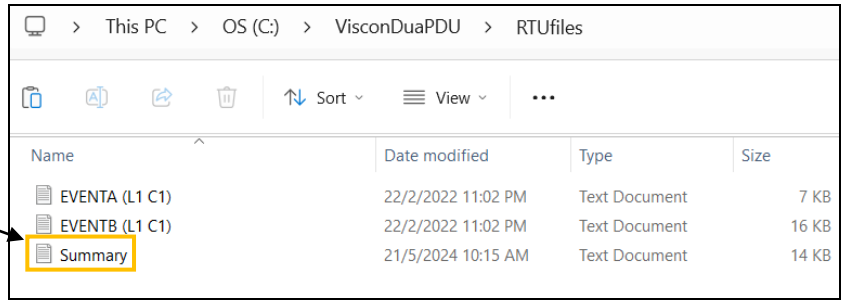
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3. Click Refresh to request settings information from the device.
4. Then click Save to save the summary of settings into PDU local machine.



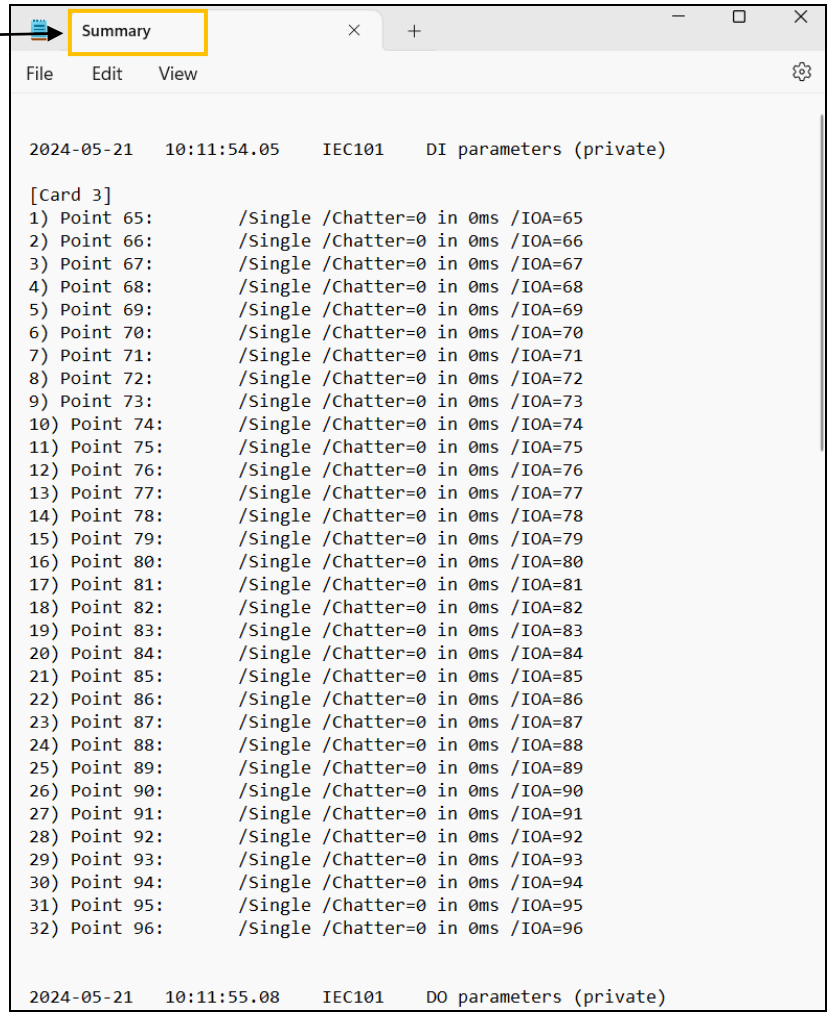
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5. The Summary file is downloaded and located at C:\VisconDuaPDU\RTUfiles



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6. Open the Summary files using Notepad application to verify the content exists.



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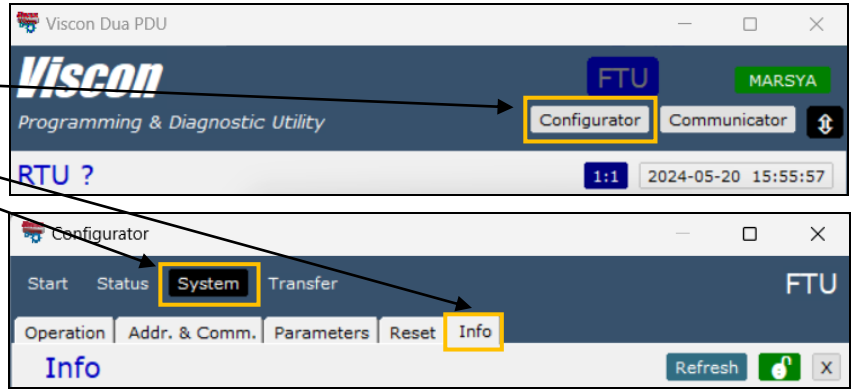
3.3 Upload Device's Firmware

3.3.1 Objectives

To demonstrate the ability of the PDU to transfer firmware file from PDU to the device.

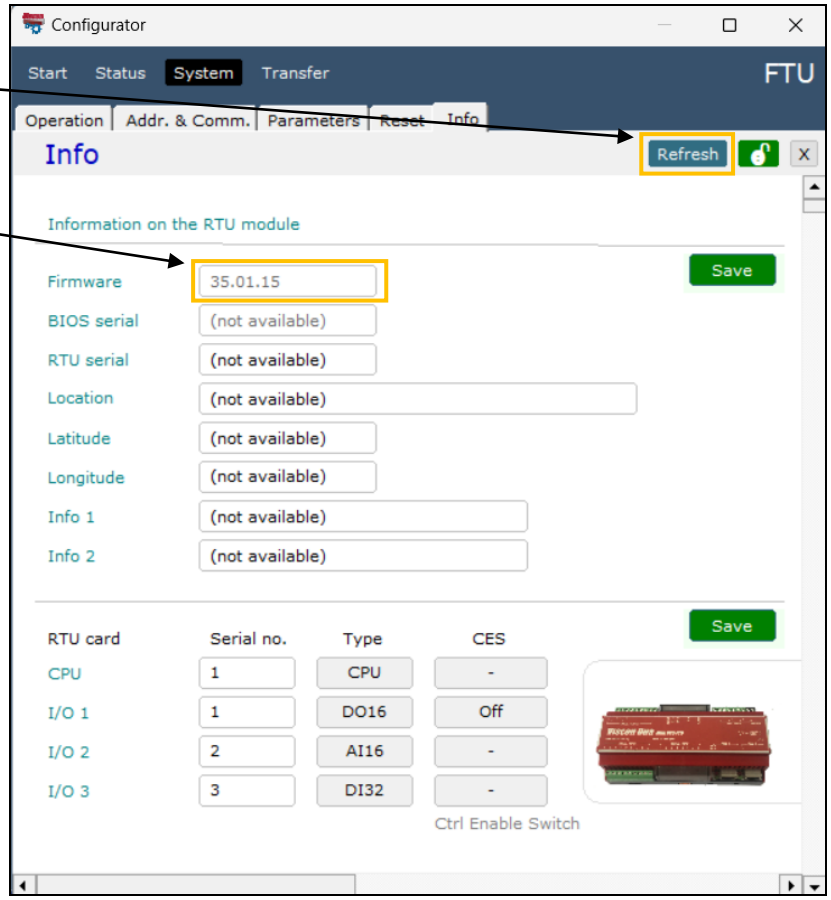
3.3.2 Procedure

1. Set PDU to communicate with the device.
2. Select Configurator > System > Info.



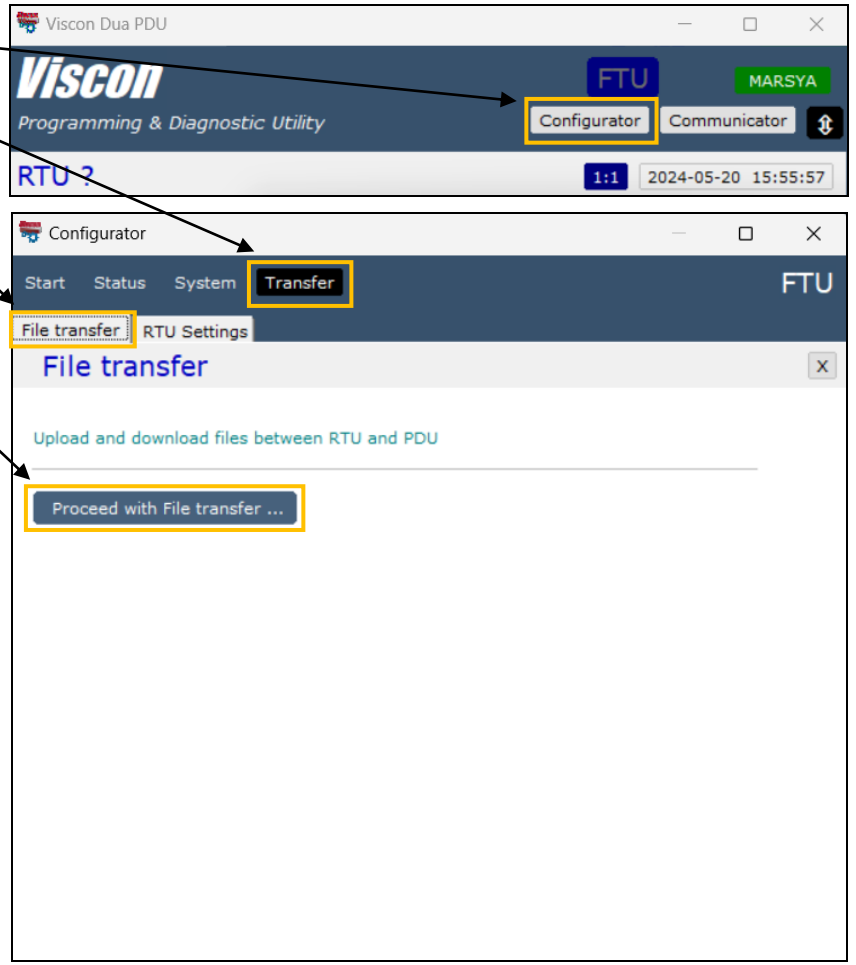
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3. Click on Refresh button to retrieve firmware version on the device.
4. Firmware version of the device will display on the respective page.



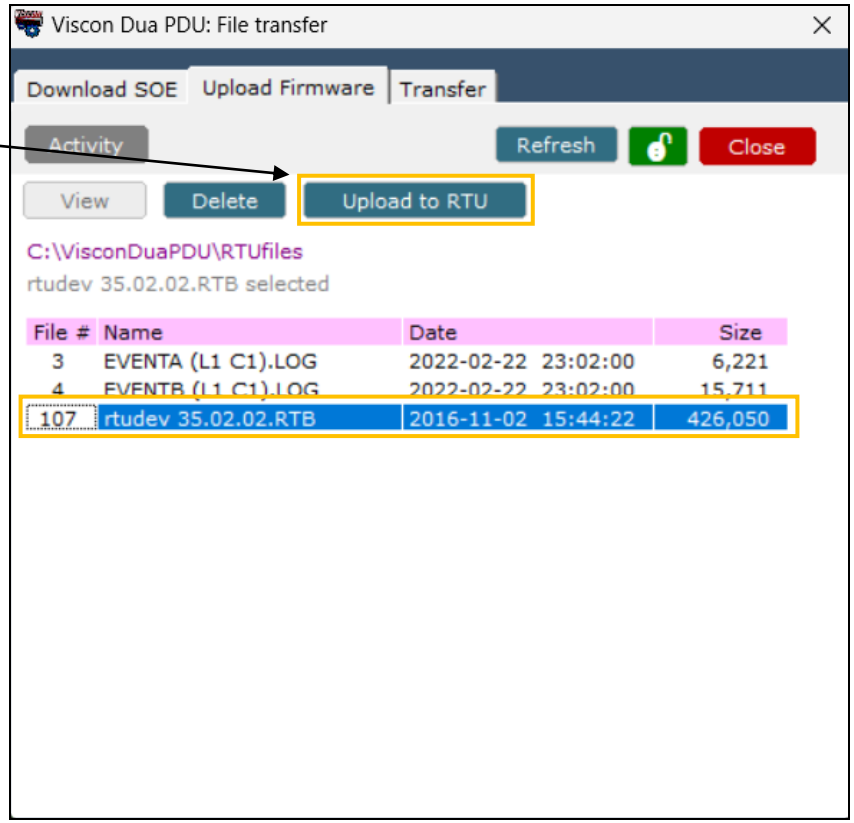
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5. Select Configurator > Transfer > File Transfer.
6. Click on Proceed with File transfer.



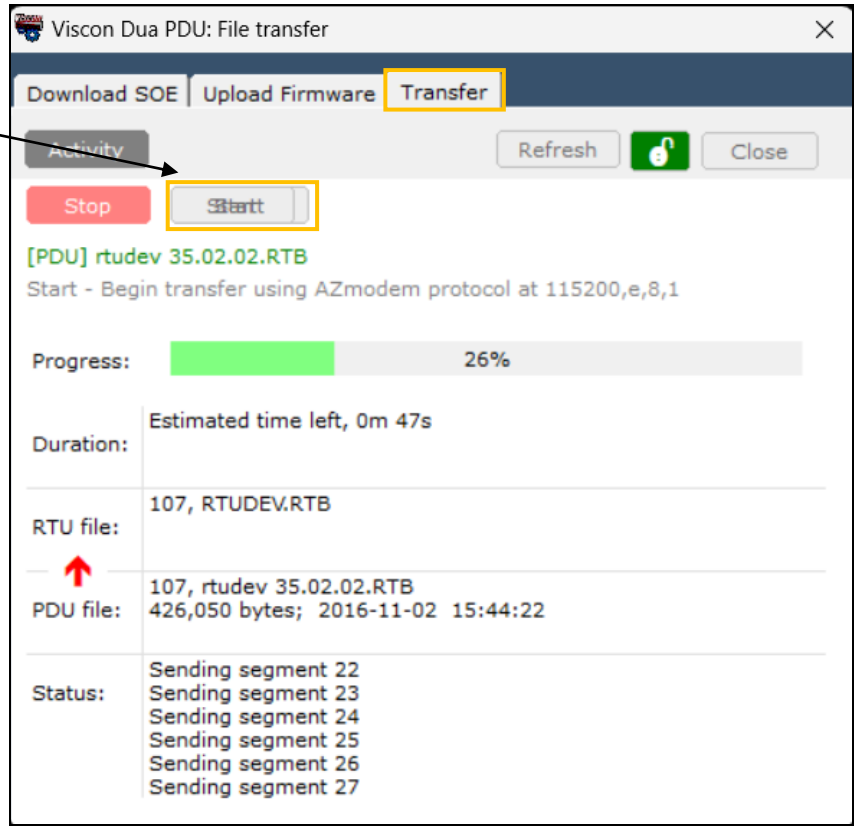
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7. In PDU files tab, select an intended firmware file (rtudev.rtb) to upload.
8. Then click Upload to RTU.



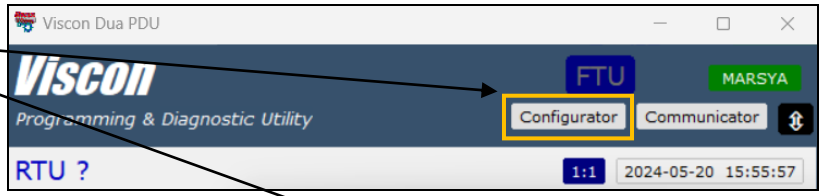
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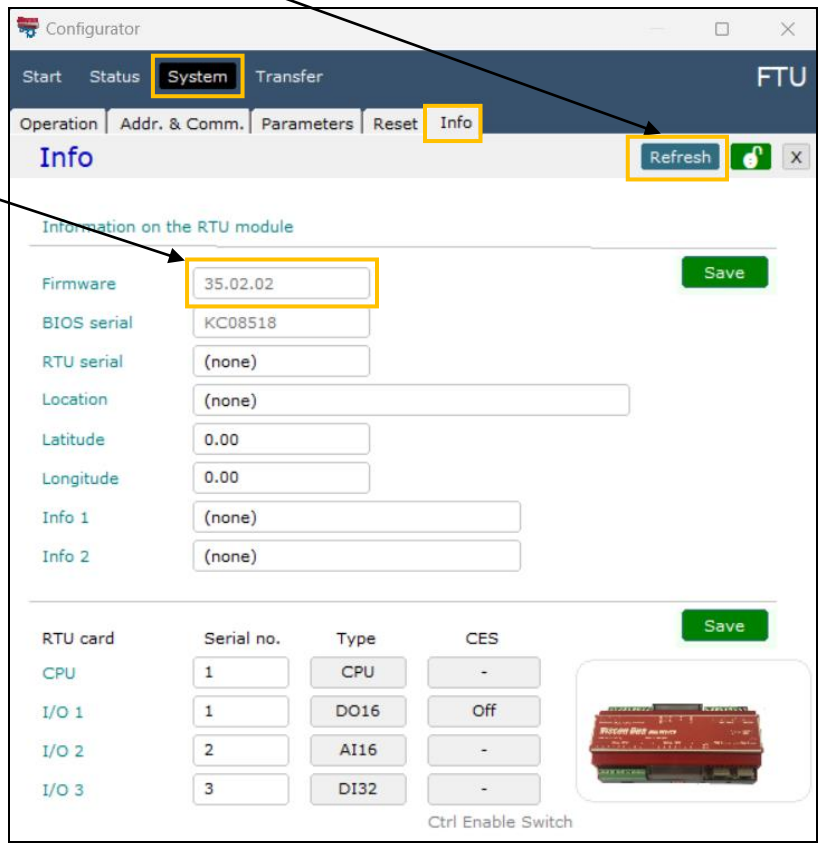
9. In the Transfer tab, click Start to start upload the firmware file into the device.
10. Wait the progress until 100%.
11. The device will automatically restart to load with new firmware.



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12. Select Configurator > System > Info.
13. Click on Refresh button to retrieve firmware version on the device.
14. New firmware version of the device will display on the respective page.





[END OF DOCUMENT]

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